Seminario Universidad de Málaga

La historia de la interleaving distance

Un camino desde la homología en $\mathbb{R}^3$ hasta sus aplicaciones

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Abstract

Empezaremos nuestra charla definiendo la interleaving distance como una métrica entre módulos persistentes. Después, para obtener un marco más general, reformularemos nuestros términos desde un punto de vista categórico. Finalizaremos con las distintas aplicaciones de la interleaving distance en estructuras como los árboles filogenéticos, merge trees, series de tiempo y grafos. Este camino estará lleno de ejemplos y referencias históricas que nos adentrarán en el maravilloso mundo del análisis topológico de datos.

Let us measure the topology of a torus using persistent homology

Consider the standard height function on the torus $h : X \to \mathbb{R}$. Define $X_a = h^{-1}(-\infty, a]$

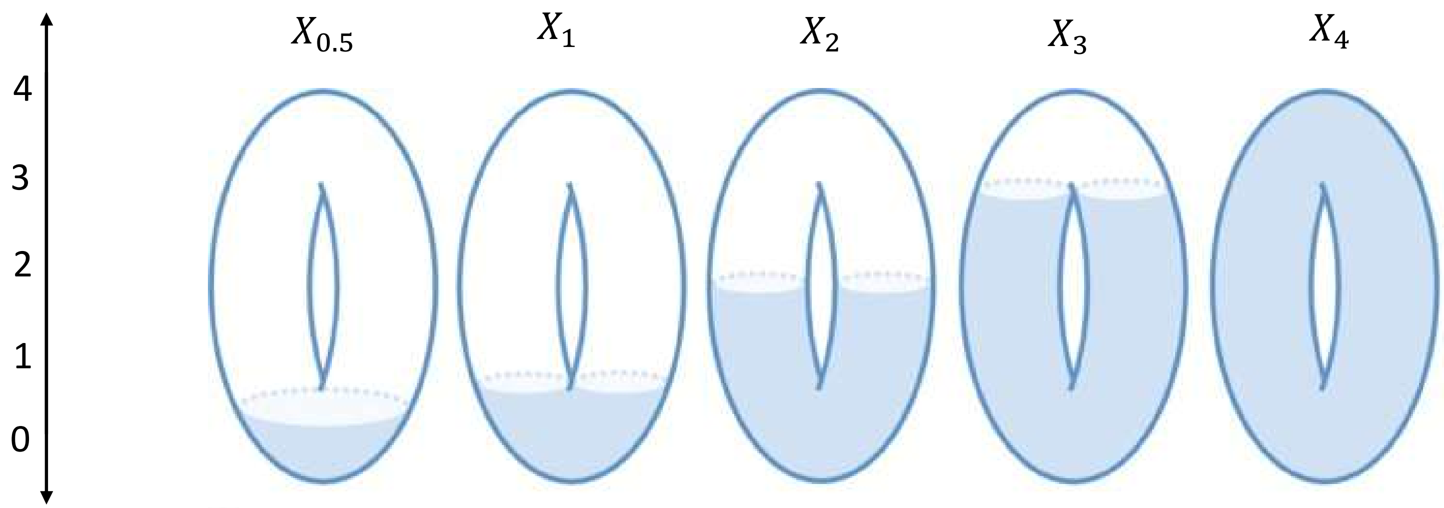
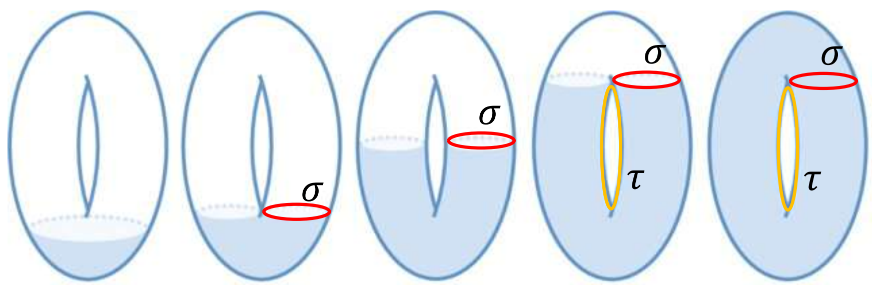


Image modified from : Curry, J. M. (2015). Topological data analysis and cosheaves. *Japan Journal of Industrial and Applied Mathematics*, *32*(2), 333–371. https://doi.org/10.1007/s13160-015-0173-9

Barcode and persistence diagram of the first homology of a torus

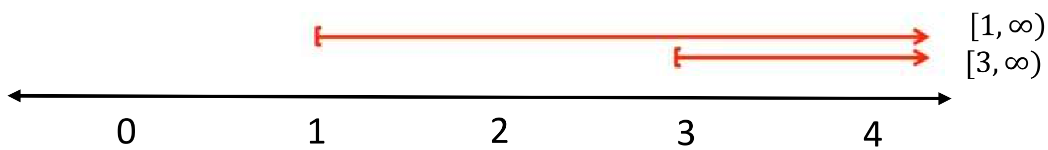
For each a take the first homology of the subspace X_a, i.e. $H_1(X_a)$.

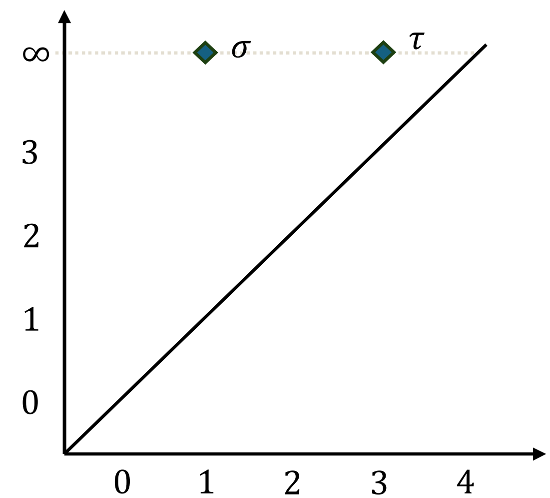
The inclusions $X_a \subset X_b$ for $a \leq b$ induces the canonical morphism $H_1(X_a) \hookrightarrow H_1(X_b)$.



H_1 $\quad [\,] \longrightarrow [\sigma] \longrightarrow [\sigma] \longrightarrow [\sigma, \tau] \longrightarrow [\sigma, \tau]$

Barcode





Persistence diagram

This notion was first introduced by Edelsbrunner, Letscher, and Zomorodian in 2002 in their article “Topological persistence and simplification”.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Persistence modules, a family of modules indexed by real numbers

We can define this chain

$$
\begin{array}{ccccccccc}
[\,] & \longrightarrow & [\sigma] & \longrightarrow & [\sigma] & \longrightarrow & [\sigma,\tau] & \longrightarrow & [\sigma,\tau] \\
0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}^2 & \longrightarrow & \mathbb{Z}^2 \\
F_0 & \longrightarrow & F_1 & \longrightarrow & F_2 & \longrightarrow & F_3 & \longrightarrow & F_4
\end{array}
$$

at algebraic level directly, without the need for an underlying functional setting.

In 2004, Zomorodian and Carlsson introduce the concept of *persistence module*

A persistence module is a family $\{F_\alpha\}_{\alpha\in\mathbb{Z}\ (A\subset\mathbb{R})}$ of modules over a same commutative ring R, together with a family of homomorphisms $\left\{f_\alpha^\beta\colon F_\alpha \to F_\beta\right\}_{\alpha\leq\beta\in\mathbb{Z}\ (A\subset\mathbb{R})}$ such that

$$f_\alpha^\gamma = f_\beta^\gamma \circ f_\alpha^\beta \text{ for } \alpha\leq\beta\leq\gamma\ ,\quad f_\alpha^\alpha = id_{F_\alpha}$$

Persistence modules are uniquely decomposed in interval modules

Let $\mathrm{I} = [\alpha_i, \infty)$ be an interval in $\mathbb{R}$. Let us define the interval $\mathbb{Z}$-module $\{\chi[\alpha_i, \infty)_a\}_{a \in I}$

$$\chi[\alpha_i, \infty)_a = \begin{cases} \mathbb{Z} \text{ if } a \in [\alpha_i, \infty) \\ 0 \text{ else} \end{cases} \qquad \mathrm{f}_\mathrm{a}^\mathrm{b}: \chi[\alpha_i, \infty)_a \to \chi[\alpha_i, \infty)_b = \begin{cases} id_{\mathbb{Z}} \text{ if } a, b \in [\alpha_i, \infty) \\ 0 \text{ else} \end{cases}$$

Do the same for $\mathrm{I} = [\beta_j, \gamma_j)$

Theorem: There is a correspondence of categories between the category of persistence modules of finite type over R and the category of finitely generated non-negatively graded modules over $R[t]$.

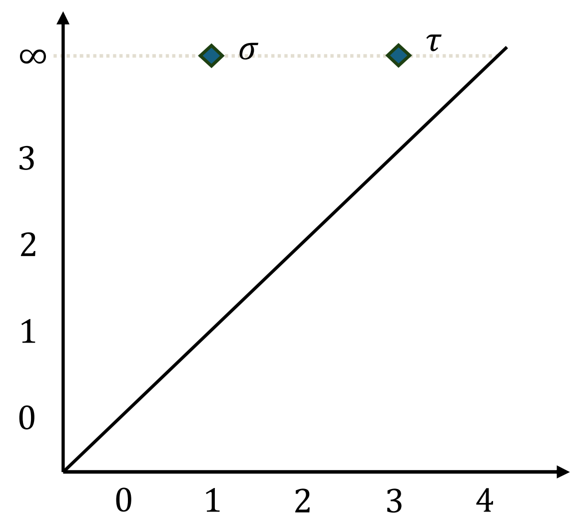
$$\hat{F} \cong \bigoplus_{i=1}^{n} \sum_i t^{\alpha_i} R[t] \oplus \bigoplus_{j=1}^{m} \sum_j t^{\beta_i} R[t]/(t^{\gamma_j}) \Rightarrow F \cong \bigoplus_{i=1}^{n} \sum_i \chi[\alpha_i, \infty) \oplus \bigoplus_{j=1}^{m} \sum_j \chi[\beta_j, \gamma_j)$$

Definition: A persistence modules$\{F_\alpha\}_{\alpha \in A \subset \mathbb{R}}$ that are *finite type* i.e. each F_α is a finitely generated R-module, and the maps f_i are isomorphisms for $i \geq m$ for some integer m.

Interval decomposition gives rise to persistence diagrams

$$
\begin{array}{lccccccccccc}
F & & [\,] & \longrightarrow & [\sigma] & \longrightarrow & [\sigma] & \longrightarrow & [\sigma,\tau] & \longrightarrow & [\sigma,\tau] & \\
F & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}^2 & \longrightarrow & \mathbb{Z}^2 & \longrightarrow \\
\chi[1,\infty) & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow \\
\chi[3,\infty) & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow \\
 & \longleftrightarrow & 0 & & 1 & & 2 & & 3 & & 4 &
\end{array}
$$

$$F \cong \chi[1,\infty) \oplus \chi[3,\infty)$$

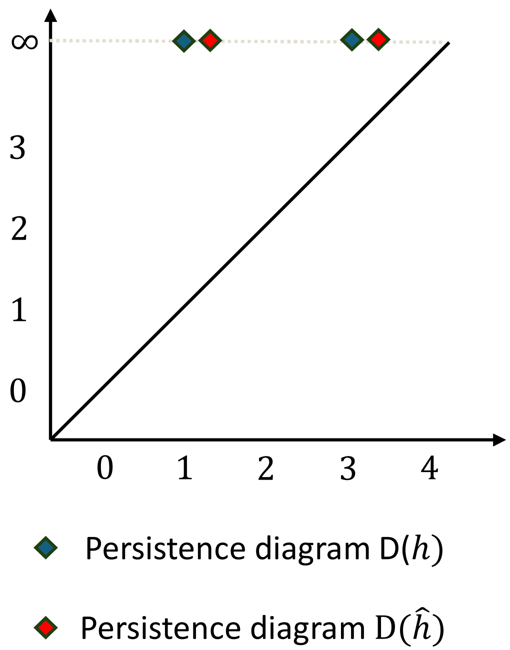


Persistence diagram

Stability of the persistence diagrams against perturbations

Consider a new height function $\hat{h}\colon X \to \mathbb{R}$ given by $\hat{h}(x) \coloneqq h(x) + \epsilon$ for $\epsilon > 0$.

Take $X_a^h = h^{-1}(-\infty, a)$ and $X_a^{\hat{h}} = \hat{h}^{-1}(-\infty, a)$. Then $X_{a+\epsilon}^{\hat{h}} = X_a^h$



Is there a relationship of the perturbation of h with the corresponding persistence diagrams of $D(h)$ and $D(\hat{h})$?

Spoiler alert: Yes

The bottleneck distance: A measure of similarity between diagrams

We can consider $D(f)$ as a multiset of points (a_i, a_j) in the extended plane $\overline{\mathbb{R}}^2$

Let P and Q be multisets in $\overline{\mathbb{R}}^2$. A partial matching between P and Q is a subset of $P \times Q$ such that

- Every point $p \in P$ is matched at most, once. So, there is at most one $q \in Q$ such that $(p, q) \in M$.
- Every point $q \in Q$ is matched at most, once. So, there is at most one $p \in P$ such that $(p, q) \in M$.

We note $M: P \leftrightarrow Q$ if M is a partial matching between P and Q. The *bottleneck cost* of M is

$$c(M) = \max\left\{\sup_{\mathrm{p,q} \in M} \left|\left|p - q\right|\right|_\infty, \sup_{s \in P \sqcup Q, unmatched} \frac{\left|s_y - s_x\right|}{2}\right\}$$

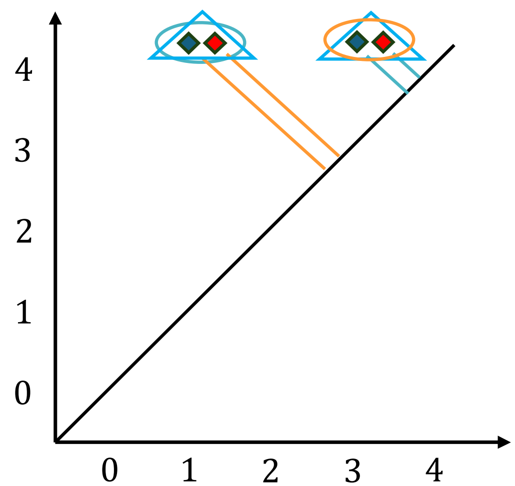
The *bottleneck distance* between P and Q is the smallest possible bottleneck cost. Then

$$\mathrm{d}_B(P, Q) = \min_{M: P \leftrightarrow Q} c(M)$$

Calculating the bottleneck distance between $D(h)$ and $D(\hat{h})$

$$\|h - \hat{h}\|_\infty = \frac{1}{2} = \epsilon$$

$$\mathrm{d}_B\big(D(h), D(\hat{h})\big) = \min_{M: D(h) \leftrightarrow D(\hat{h})} \underbrace{\max\left\{\sup_{\mathrm{p,q} \in M} \|p - q\|_\infty, \sup_{s \in P \sqcup Q, unmatched} \frac{|s_y - s_x|}{2}\right\}}_{c(M)}$$



◆ Persistence diagram of $D(h)$ {{ [1,5), [3,5) }}

◆ Persistence diagram of $D(\hat{h})$ {{ $[1+\epsilon, 5), [3+\epsilon, 5)$ }}

$$\mathrm{d}_B\big(D(h), D(\hat{h})\big) = min \left\{ \begin{array}{c} \text{Matching 1} \\ c(M_1) = \max\left\{\epsilon, \max\left\{2 + \frac{\epsilon}{2}, 2\right\}\right\} = 2 + \frac{\epsilon}{2} \\ \text{Matching 2} \\ c(M_2) = \max\left\{\epsilon, \max\left\{1 + \frac{\epsilon}{2}, 1\right\}\right\} = 1 + \frac{\epsilon}{2} \\ \text{Matching 3} \\ c(M_3) = \max\{\max\{\epsilon, \epsilon\}\} = \epsilon \end{array} \right\} = \epsilon$$

Stability theorem for persistence diagrams

Theorem: Let X be a *triangulable* space with *continuous tame* functions $f, g : X \to \mathbb{R}$. Then the persistence diagrams satisfy

$$d_B\big(D(f), D(g)\big) \leq \|f - g\|_\infty$$

This theorem was proved by Edelsbrunner and Harer in 2006 in their article "Stability of persistence diagrams".

Definition: A topological space X is triangulable if there is a (finite) simplicial complex with homeomorphic underlying space

Definition: A function $f : X \to \mathbb{R}$ is tame if it has a finite number of homological critical values and the homology groups $H_k(f^{-1}(-\infty, a])$ are finite-dimensional for all $k \in \mathbb{Z}$ and $a \in \mathbb{R}$.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Stability theorem: working on the algebraic level to remove some limitations

Theorem: Let X be a *triangulable* space with *continuous tame* functions $f, g : X \to \mathbb{R}$. Then the persistence diagrams satisfy

$$d_B\big(D(f), D(g)\big) \leq \|f - g\|_\infty$$

Limitations : (1) X is triangulable, (2) f and g are continuous, (3) tame and (4) defined over the same X.

In 2008, Chazal, Cohen-Steiner, Glisse, Guibas and Oudot removed continuity and triangulability conditions, the tameness condition is relaxed, and functions can be defined over different topological spaces.

To achieve this result, we drop the functional setting and work at algebraic level directly.

Theorem: Take $X_a^f = f^{-1}(-\infty, a)$. Let $\mathrm{F(a)} = H_k(X_a^f)$ and $\mathrm{G(a)} = H_k(X_a^g)$ be two *tame* persistence modules ($\dim F(a) < \infty$). If F and G are strongly ϵ-interleaved, then

$$d_B\big(D(f), D(g)\big) \leq \epsilon$$

Interleaving of persistence modules

Two persistence modules F and G are said to be strongly ε-interleaved if there exist two families of homomorphisms

$$\{\varphi(a): F(a) \to G(a+\varepsilon)\} \text{ and } \{\psi(a): G(a) \to F(a+\varepsilon)\}$$

such that the following diagrams commute for all $a \in \mathbb{R}$

$$\begin{tikzcd}
F(a) \arrow[rr] \arrow[dr, "\varphi(a)"'] & & F(b) \arrow[dr, "\varphi(b)"] & \\
& G(a+\varepsilon) \arrow[rr] & & G(b+\varepsilon)
\end{tikzcd}
\qquad
\begin{tikzcd}
& F(a+\varepsilon) \arrow[rr] & & F(b+\varepsilon) \\
G(a) \arrow[ur, "\psi(a)"] \arrow[rr] & & G(b) \arrow[ur, "\psi(b)"'] &
\end{tikzcd}$$

$$\begin{tikzcd}
F(a) \arrow[rr] \arrow[dr, "\varphi(a)"'] & & F(a+2\varepsilon) \\
& G(a+\varepsilon) \arrow[ur, "\psi(a+\varepsilon)"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
& F(a+\varepsilon) \arrow[dr, "\varphi(a+\varepsilon)"] & \\
G(a) \arrow[ur, "\psi(a)"] \arrow[rr] & & G(a+2\varepsilon)
\end{tikzcd}$$

The isometry theorem: Bottleneck and interleaving distances are the same

We define the *interleaving distance* between the modules F and G by

$$d_I(F,G) \coloneqq \inf\{\epsilon \geq 0 \mid F, G \text{ are } \epsilon\text{−interleaved}\}$$

Note that d_I is a pseudometric . e.g $d_I(\chi[0,1], \chi(0,1]) = 0$ even though they are not isomorphic.

Theorem: Take F and G two *tame* persistence modules then

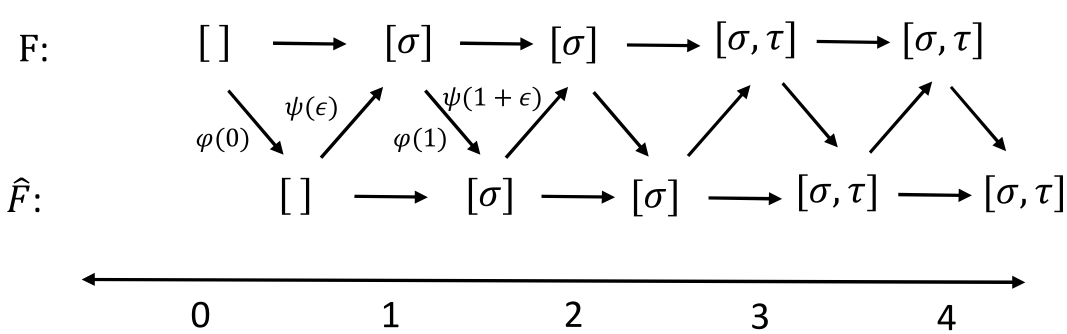
$$d_B(F,G) = d_I(F,G)$$

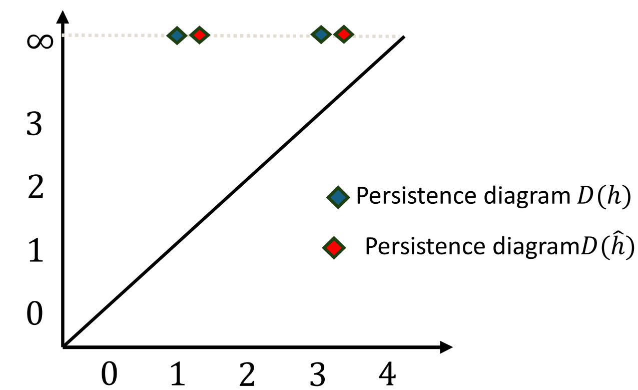
Proof: $d_B(F,G) \leq d_I(F,G)$ from the stability theorem via interpolation theorem. Create a series of F_s that is strongly s-interleaved with F and strongly $(\epsilon - s)$-interleaved with G

$d_I(F,G) \leq d_B(F,G)$ using F, G the decomposition in interval modules.
(Proved by Lesnick in 2011 in his article the Theory of the Interleaving Distance on Multidimensional Persistence Modules)

Calculating the interleaving distance between h and $\hat{h}$

Let us take the persistence modules $\mathrm{F(a)} = H_1\left(X_a^h\right)$, $\hat{F}\mathrm{(a)} = H_1\left(X_a^{\hat{h}}\right)$





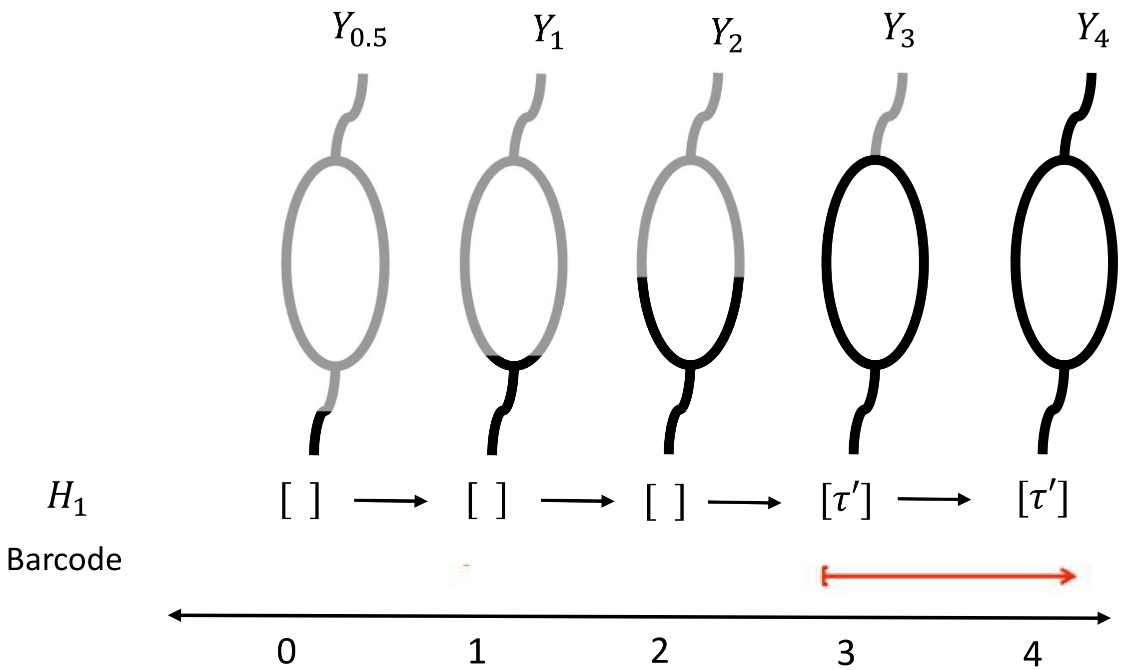
Recall that $\left\|h - \hat{h}\right\|_\infty = \frac{1}{2} = \epsilon$. Then, the sublevel sets filtrations are nested as follows, $X_a^h \subseteq X_{a+\epsilon}^{\hat{h}} \subseteq X_{a+2\epsilon}^h$ for all $a \in \mathbb{R}$. This nesting makes F and $\hat{F}$ ϵ-interleaved. Therefore

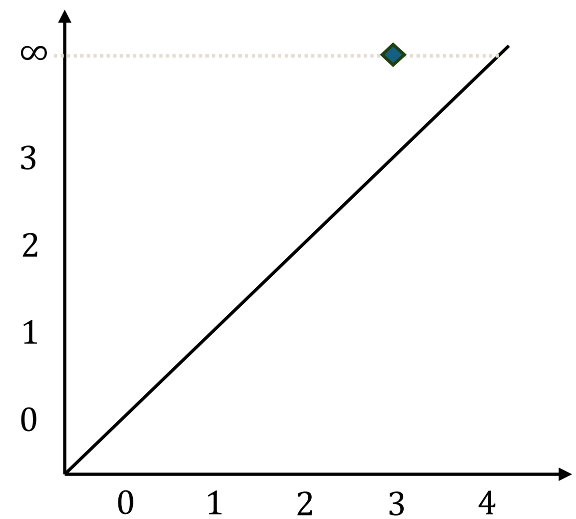
$$d_B\left(D(h), D\left(\hat{h}\right)\right) = \frac{1}{2} = d_I(F, \hat{F})$$

The topology of the Reeb graph of the torus using persistent homology

Consider Y the Reeb graph of the torus induced by h and use the same height function $h'\colon Y \to \mathbb{R}$.

Now we can we compare it with the persistent homology of the torus.

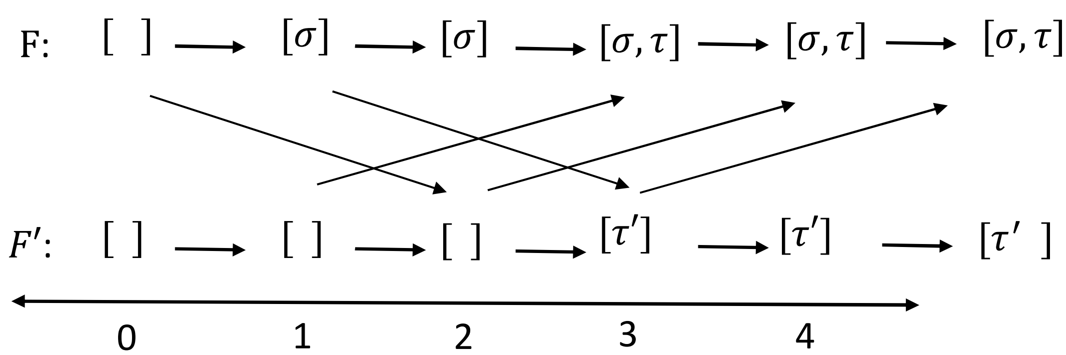


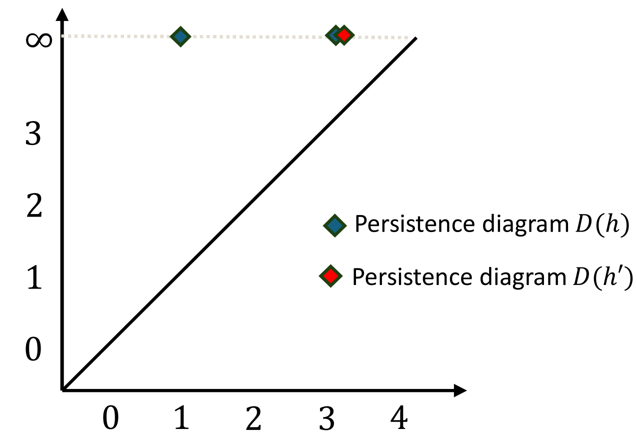


Persistence diagram

Calculating the interleaving distance of our examples

Let us take the persistence modules $\mathrm{F(a)} = H_1\left(X_a^h\right)$, $\mathrm{F'(a)} = H_1\left(Y_a^{h'}\right)$





$\mathrm{d}_B(D(h), D(h')) = 2$

$$\begin{tikzcd} {[\sigma]} \arrow[rr] \arrow[dr] & & {[\sigma,\tau]} \\ & {[\ ]} \arrow[ur] & \end{tikzcd} \qquad \begin{tikzcd} {[\sigma]} \arrow[rr] \arrow[dr] & & {[\sigma,\tau]} \\ & {[\tau']} \arrow[ur] & \end{tikzcd}$$

F and F' are not ϵ-interleaved. The diagrams doe not commute.

$$\mathrm{d}_B(D(h), D(h')) = \infty = d_I(F, F')$$

Categorification of persistent homology

In 2014, Bubenik and Scott redevelop topological persistence from a categorical point of view.

A **persistent module** F is a functors from $\mathbb{R}$ (poset) to Vec. (Actually, any category $\boldsymbol{C}$)

For $b \geq 0$, define $\Omega_b : (\mathbb{R}, \leq) \to (\mathbb{R}, \leq)$ to be the shift functor given by $\Omega_b(a) = a + b$, and
Define $\eta_b : Id(\mathbb{R}, \leq) \Rightarrow \Omega_b$ to be the natural transformation given by $\eta_b(a) : a \leq a + b$.

An ***ϵ-interleaving*** of F and G consists of natural transformations

$$\varphi : F \Rightarrow G\Omega_\epsilon \text{ and } \psi : G \Rightarrow F\Omega_\epsilon$$

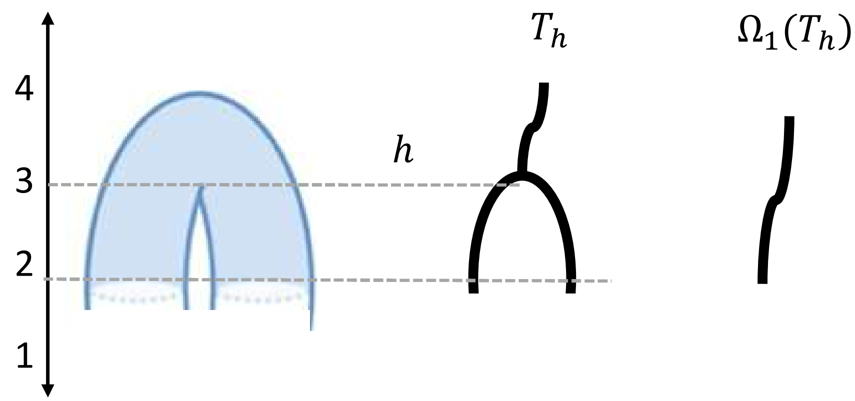
$$\begin{tikzcd}
F(a) \arrow[rr] \arrow[dr, "\varphi(a)"'] & & F(b) \arrow[dr, "\varphi(b)"] & \\
& G(a+\varepsilon) \arrow[rr] & & G(b+\varepsilon)
\end{tikzcd}
\qquad
\begin{tikzcd}
& F(a+\varepsilon) \arrow[rr] & & F(b+\varepsilon) \\
G(a) \arrow[ur, "\psi(a)"] \arrow[rr] & & G(b) \arrow[ur, "\psi(b)"'] &
\end{tikzcd}$$

such that

$$(\psi\Omega_\epsilon)\varphi = F\eta_{2\epsilon} \text{ and } (\varphi\Omega_\epsilon)\psi = G\eta_{2\epsilon}$$

$$\begin{tikzcd}
F(a) \arrow[rr] \arrow[dr, "\varphi(a)"'] & & F(a+2\varepsilon) \\
& G(a+\varepsilon) \arrow[ur, "\psi(a+\varepsilon)"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
& F(a+\varepsilon) \arrow[dr, "\varphi(a+\varepsilon)"] & \\
G(a) \arrow[ur, "\psi(a)"] \arrow[rr] & & G(a+2\varepsilon)
\end{tikzcd}$$

Merge trees as persistent modules



The merge tree T_h captures the connected components among the sublevels of (X, h)

$$T_h\colon \mathbb{R} \to Set$$
$$a \mapsto X_a = h^{-1}(-\infty, a]/\sim$$

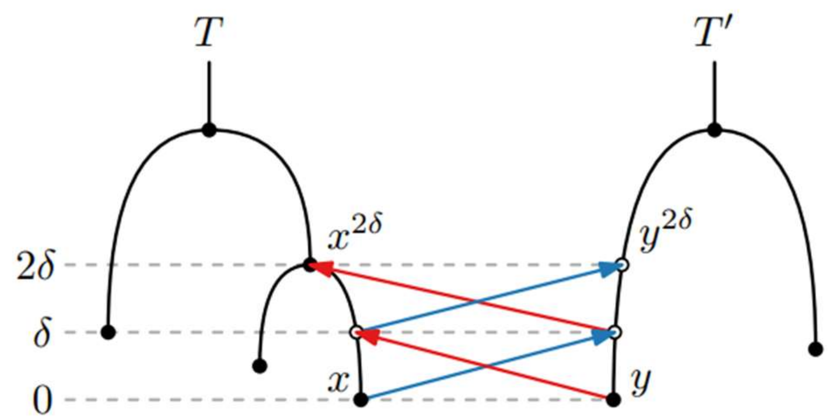
The shift ϵ of T_h is given by

$$\Omega_\epsilon(T_h)(a) = T_h(a+\epsilon)$$

An ***ϵ-interleaving*** of T and T' consists of maps

$$\varphi\colon T \to \Omega_\epsilon(T') \qquad \psi\colon T' \to \Omega_\epsilon(T)$$

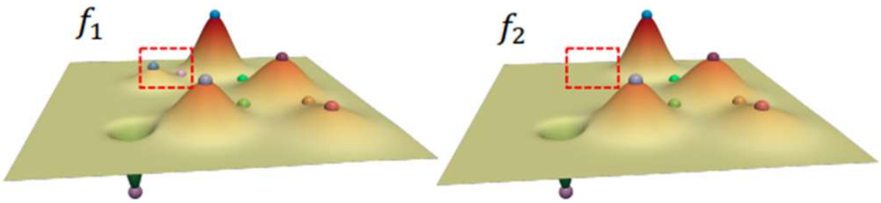
that are compatible

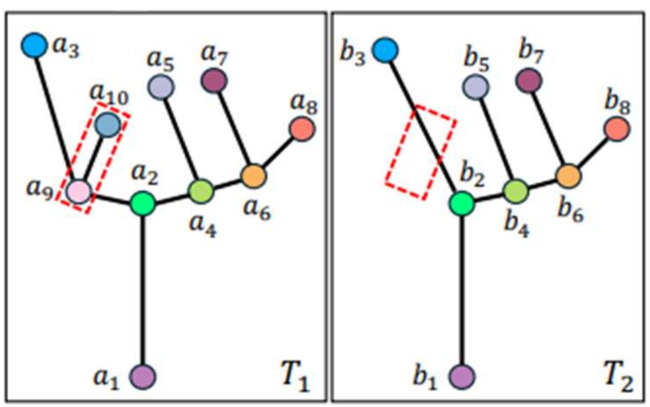


The distance is given by the infimum over all ϵ
That make T and T' interleaved

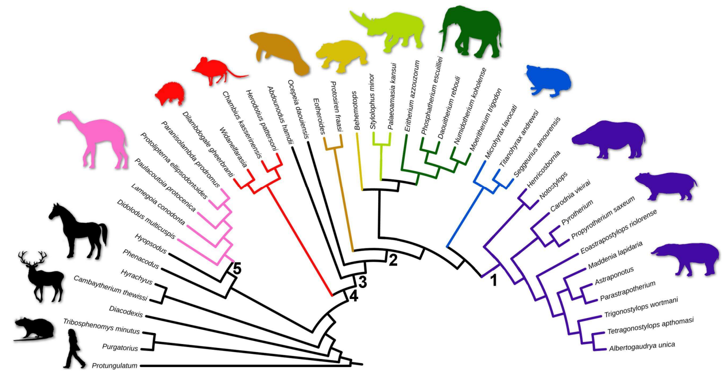
Merge trees are topological descriptor for data with a hierarchical component

Scalar fields

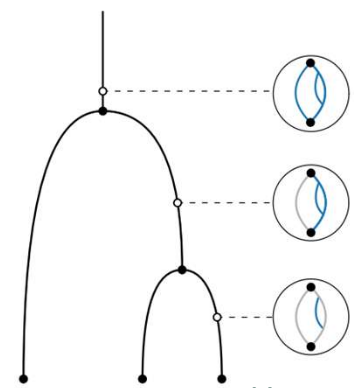




Phylogenetic trees

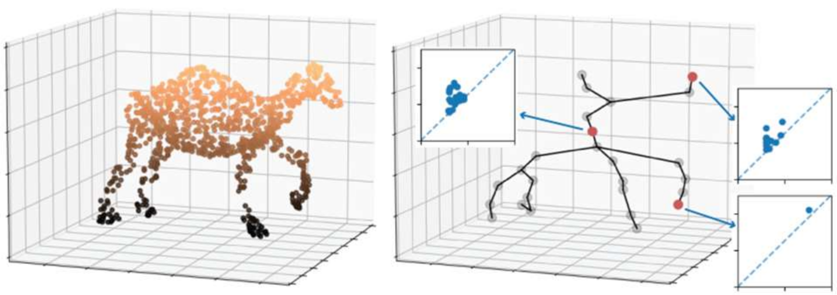


A river network

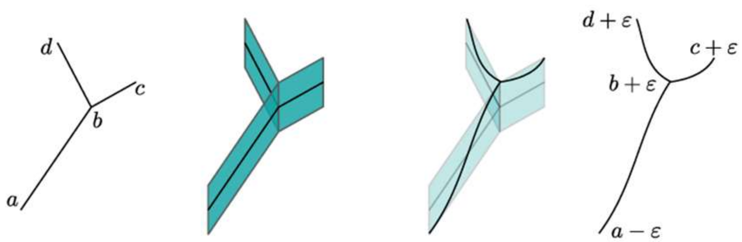


More structures view as persistence modules

Reeb graphs



(a) Point Cloud (b) Decorated Reeb Graph



1. Image from Curry, J., Mio, W., Needham, T., Okutan, O. B., & Russold, F. (2023). Stability and approximations for decorated reeb spaces. arXiv preprint arXiv:2312.01982.
2. Image from De Silva, V., Munch, E., & Patel, A. (2016). Categorified reeb graphs. Discrete & Computational Geometry, 55(4), 854-906.

Multiparameter diagrams

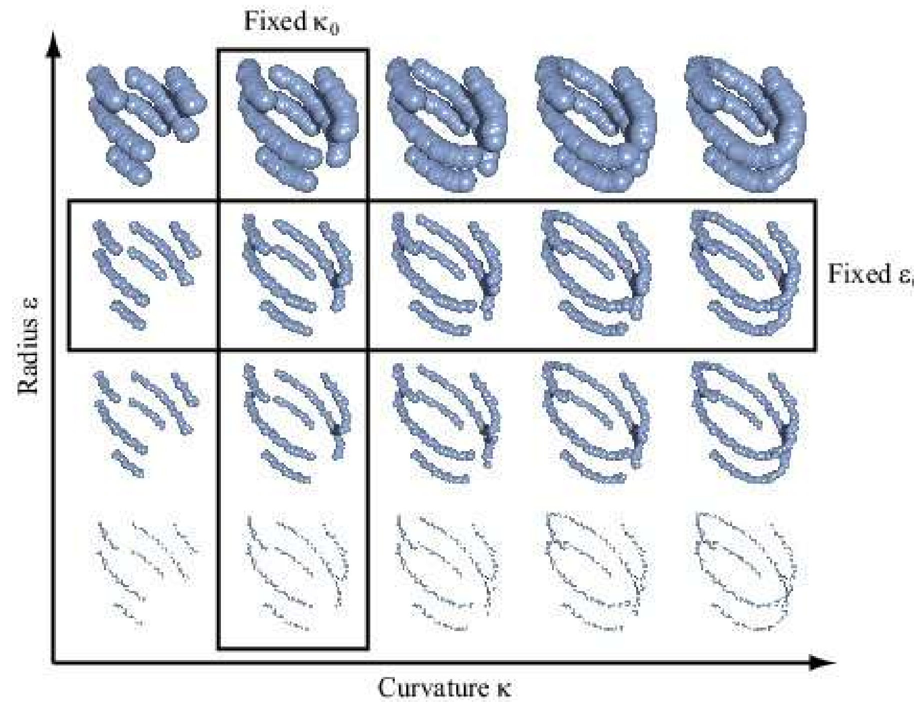


Image from Carlsson, G., & Zomorodian, A. (2007, June). The theory of multidimensional persistence. In Proceedings of the twenty-third annual symposium on Computational geometry (pp. 184-193).

Time series as persistent modules

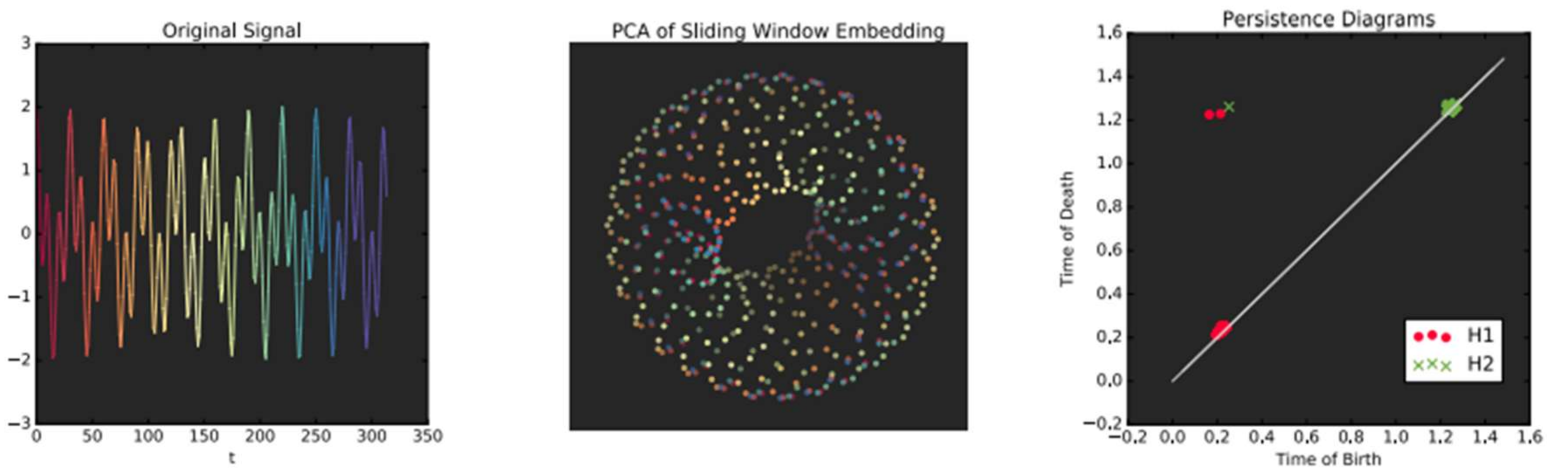


Image modified from Tralie, C. J., & Perea, J. A. (2018). (Quasi) periodicity quantification in video data, using topology. *SIAM Journal on Imaging Sciences*, *11*(2), 1049-1077.

Graph networks as persistent modules

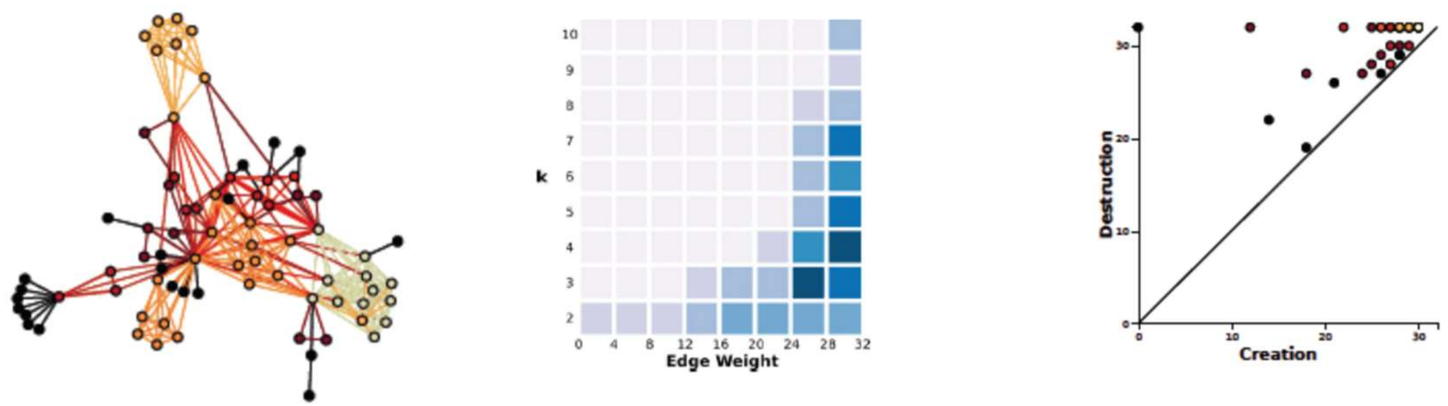
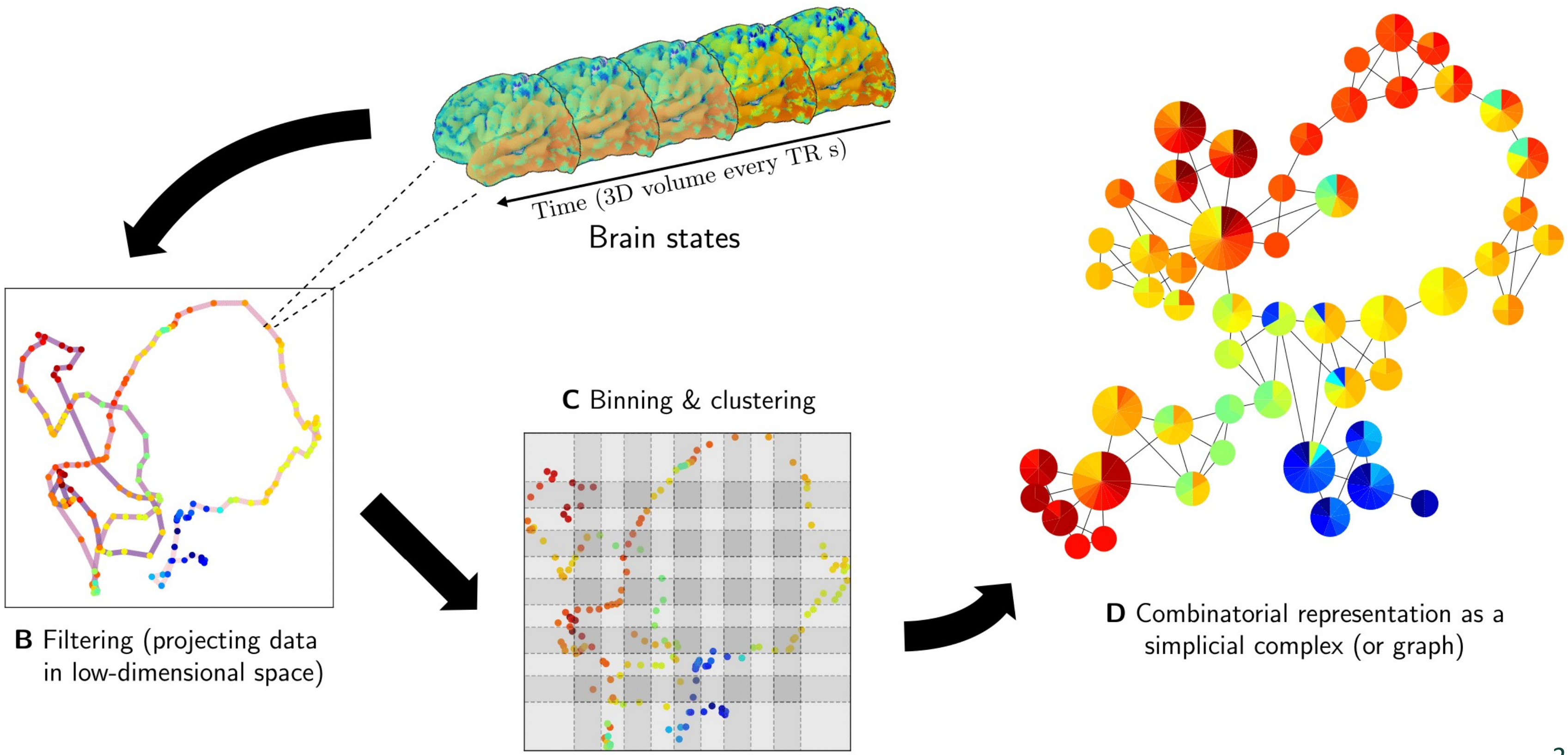


Image modified from Rieck, B., Fugacci, U., Lukasczyk, J., & Leitte, H. (2017). Clique community persistence: A topological visual analysis approach for complex networks. *IEEE transactions on visualization and computer graphics*, *24*(1), 822-831.

Mapper provides a graph capturing the topological features of the dataset

Olave, A. A., Perea, J. A., & Gómez, F. (2025). Revealing brain network dynamics during the emotional state of suspense using TDA. *Network Neuroscience*, 1–25

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Mappers as persistent modules

The mapper graph M captures the connected components inside the bins of the grid $K \subset \mathbb{R}^d$

$$M: Open(K) \to Set$$
$$\mathrm{S} \quad \mapsto \pi_0\, f^{-1}(S)$$

The shift ϵ of M is given by

$$\Omega_\epsilon(M)(S) = \; M(S^\epsilon)$$

The distance between the mappers X and Y is given by the minimum n that has induced maps

$$\varphi: X \to \Omega_\epsilon(Y) \qquad \psi: Y \to \Omega_\epsilon(X)$$

compatible with the connected components

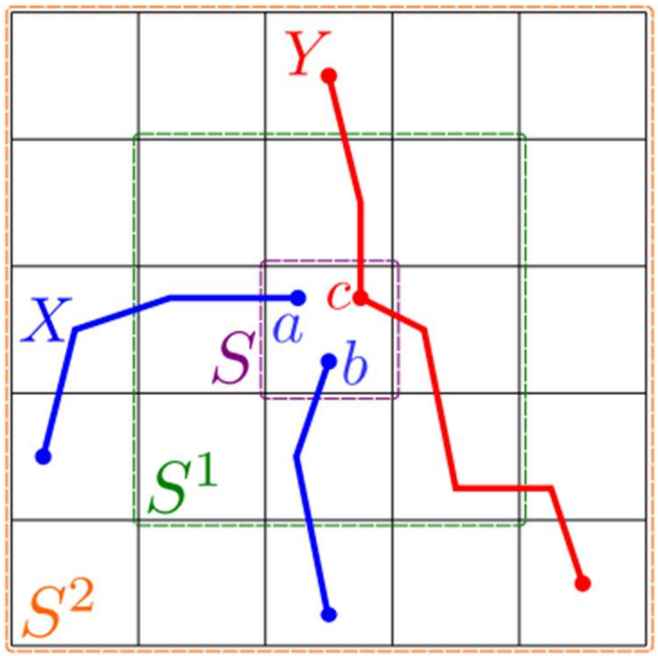

Image from Chambers, E. W., Munch, E., Percival, S., & Wang, B. (2023). Bounding the Interleaving Distance for Mapper Graphs with a Loss Function. *arXiv preprint arXiv:2307.15130*.

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! Gracias !

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